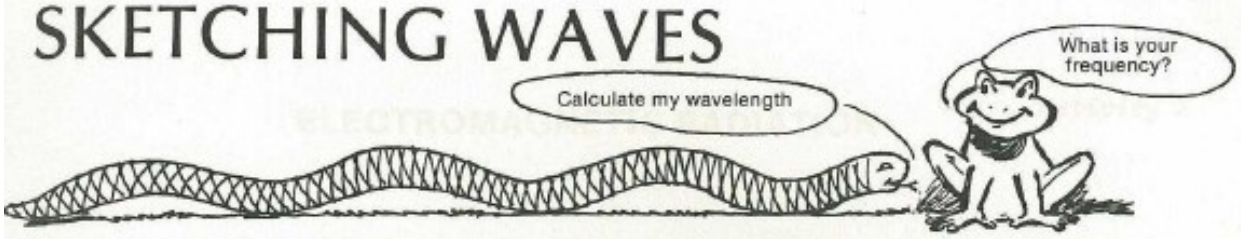
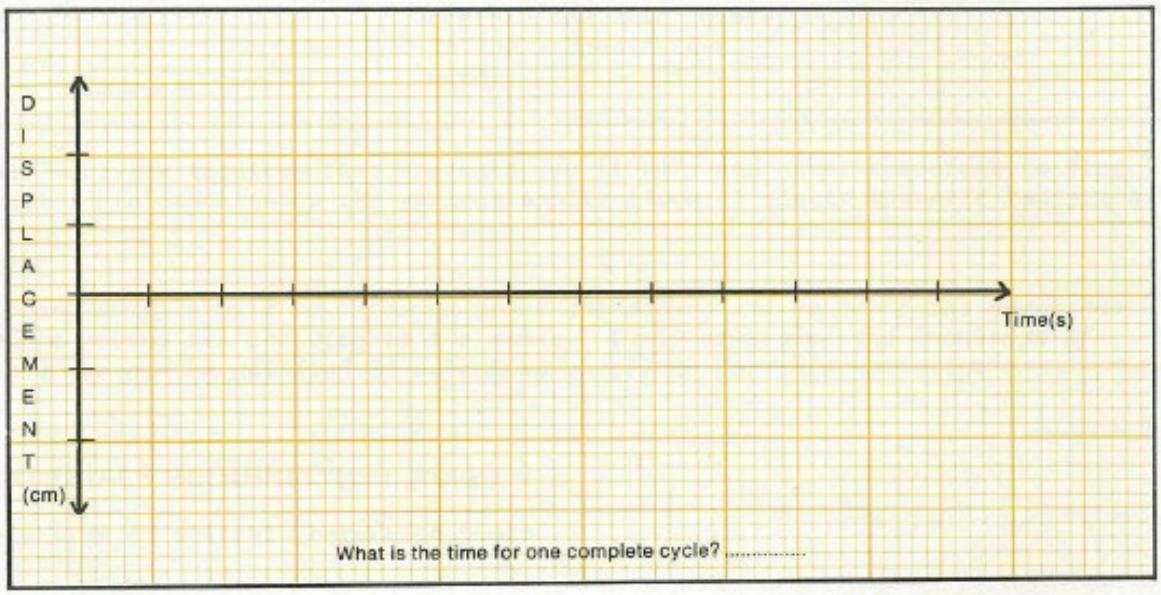
SKETCHING WAVES



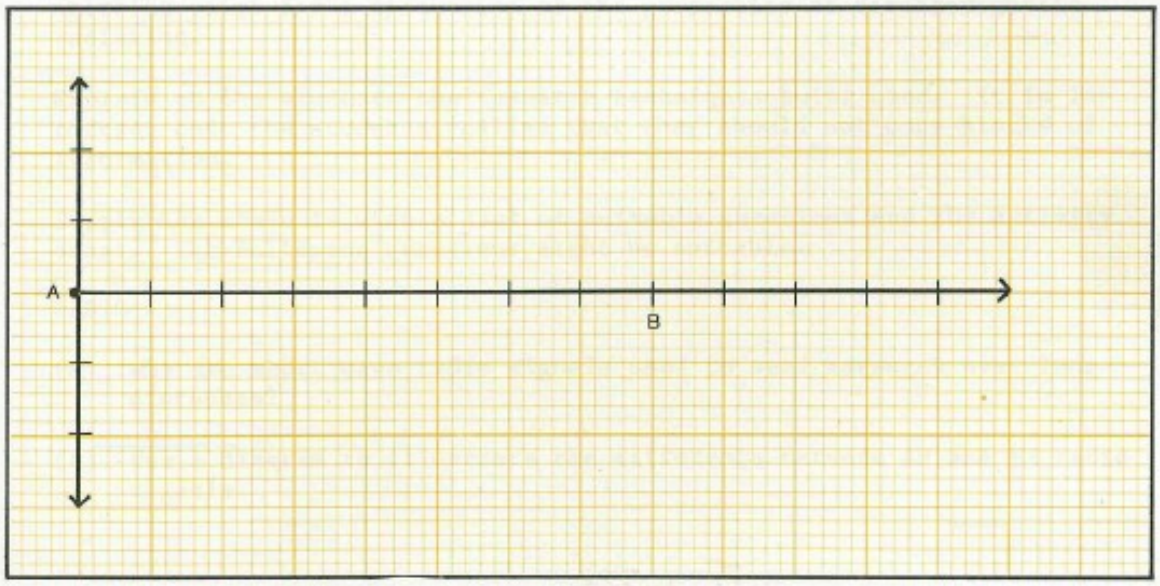
The following data have been obtained by photo-analysis of the motion of a single particle in a medium as energy in a wave form passes through the medium.

Time (s)	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2.0
Displacement from rest position (cm)	0	5	11	16	19	20	19	16	11	5	0	-5	-11	-16	-19	-20	-19	-16	-11	-5	0

Plot the data on the graph paper.



Using the piece of wire you have been given, construct the outline of a wave with a wavelength of 4 cm and an amplitude of 3 cm. Trace the outline onto the graph paper.



If the wave takes 1 second to travel from A to B, calculate the frequency.